

APPROVAL

This is to certify that this report was written by TSAFACK IVAN MARC and accepted as partial fulfilment of the requirements for the Award of a Bachelor of Technology (B. Tech) degree in Cloud computing.

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Date

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CONFIANCE

DEDICATION



AGKNOWLEDGEMENTS

I have taken efforts in this project. However, it would not have been possible without the kind support and help of many individuals and organizations. I will like to extend my sincere thanks to all of them. The accomplishment of this work required the assistance of some people. Failing to quote them individually here, they receive my most sincere thanks.

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- My family for their moral, material and financial support.
- To the ALMIGHTY GOD who gave me good health, strength, wisdom, and countless other good things for the accomplishment of this work.

PREFACE

With the aim of ensuring suitable development and providing business with a competent and competitive workforce in various fields, the Cameroonian government through the ministry of higher Education has opened the private institution of education. This giving the opportunity to private institutions, to contribute to the acquisition of an academic and professional training in adequacy with the professional world.

This created the ISTD (Higher Institute of Technology and Industrial Design) by decree N° 02/OO94 /MINESUP /DDES /ESUP of September 13 2002 and authorization of opening N° 0102/ MINESUP /DDES/ ESUP of 18 September 2002. Located in the coastal region, Wouri department, district of Douala 5Th, in the Logbessou district. The ISTD is then created in University Institute of the Coast (IUC) by decree N° 5/05156/ N/ MINESUP/ DDES /ESUP/ SAC/ ebm, and comprises to date five (5) establishments in particular;

1. School of Engineering and Applied Sciences (SEAS).

The courses taught in this school relate to the commercial, communication and management sectors. It is here that the senior managers and entrepreneurs in management and international trade are formed, with particularly African; European and Asian. All supported by language stays abroad.

➤ HND INDUSTRIAL PROGRAMS

- Electrical power systems (EPS)
- Industrial Computing and Automation (ICA)
- Telecommunication (TEL)
- Software Engineering (SWE)
- Networks and security (NWS)
- Chemical Manufacturing (CM)
- Mechanical Manufacturing (MEM)
- Building Science and Technology (BST)

➤ HND COMMERCIAL PROGRAMS

- Marketing (MKT)
- Accountancy (ACC)

- Banking and Finance (BF)
- Logistics and transport management (LTM)
- BTS PROGRAM
 - GENIE CIVIL (GC)
 - GEOMETRE TOPOGRAPH (GT)
 - BATIMENTS
 - TRAVAUX PUBLIC (TPU)
- BACHELOR OF ENGINEERING (B-Eng)
 - Computer Science Engineering
 - Electromechanical Engineering
 - Civil and Environmental Engineering
- BACHELOR OF TECHNOLOGY (B-Tech)
 - Chemical Engineering
 - Electrical and Electronics Engineering
 - Civil Engineering
 - Software Engineering
 - Mecatronics Engineering
 - Industrial automation and computing
 - Welding and Fabrication Technology
 - Thermal Engineering Technology
- BACHELOR OF HEALTH SCIENCE (B-HES)
 - Physiotherapy
 - Medical Laboratory
 - Nursing science
- LIPRO
 - Génie Civil
- MPRO
 - Genie Civil

2. The Institute of Computer Engineering of Central Africa (ICIA)

3IAC is an institution that has sub regional coverage that offers students the opportunity to train in the fields of industry and technology. Its particularity is to train the holder of the scientific BACs, in the basic sciences, for the direct integration in the third year in a computer

engineering cycle at the IUC or in the partner schools. Students have a double degree and a training program in perfect harmony with the model of foreign partners.

➤ **BTS PROGRAMS**

- Action Commerciale
- Assurance
- Banque et Finance
- Communication d'Entreprise
- Commerce International
- Gestion Logistique et Transport
- Informatique de Gestion

➤ **LICENCES PROFESSIONNELLES COMMERCIALES ET GESTION** in partnership with the University of DSCHANG:

- Assurance
- Banque
- Banque – Assurance
- Comptabilité Contrôle et Audit
- Finance – Comptabilité
- Gestion des Ressources Humaines
- Logistique et Transport
- Marketing
- Management des Opérations du Commerce International

➤ **MASTER PROFESSIONNEL COMMERCIAL** in partnership with the University of DSCHANG

- Finance-Comptabilité
- Audit Comptable et Financier
- Banque
- Fiscalité
- Management des Organisation
- Marketing-Communication
- Gestion des Ressources Humaines
- Logistique et Stratégique

➤ **MASTER ISUGA-FRANCE** in partnership with EMBA France

- EXECUTIVE MBA in partnership with ESSEC Douala

3. L'Institut d'Ingénierie Informatique d'Afrique Centrale (3IAC)

➤ **Cycle des TIC** in partnership with CCNB-DIEPPE of CANADA

- Réseautique et Sécurité
- Programmes et Analyse
- MASTER EUROPEAN in partnership with 3IL France
 - Genie Logiciel
 - Administration des Systèmes Réseaux
- MASTER PROFESSIONNEL in partnership with ENSP Yaoundé
 - Genie Civil
 - Genie Energétique et Environnement
 - Génie Industriel et Maintenance
- CYCLE INGENIEUR Information (3IL-France)
 - Pole de Recherche Innovation et Entrepreneuriat

4. International Programs of Science and Technology Innovation (PISTI)

This school offers training oriented towards preparatory classes (chemistry, physics and mathematics applied to finance); agric-food engineering; Biomedical engineering Architecture and design, etc. these programs allow for better preparation (psychology, socio – cultural, linguistic, etc.) of children who obtain a baccalaureate at a very young age so that they can integrate well once abroad.

➤ Cycle des Classes Préparatoires aux Grandes Ecoles d'Ingénieurs (CP) in partnership with the University of MAINE FRANCE:

- Classes préparatoires
- Licences Sciences et techniques
- CYCLE INGENIEUR de Génie Industriel (Polytechnique-NANCY France)

➤ INGENIEURIE BIOMEDICALE in partnership with the University of Tor Vergata (ROME II)

➤ ARCHITECTURE ET DESIGN INDUSTRIEL in partnership with the University of CAMERINO (ROME I)

5. L'Institut Supérieur des Technologies et du Design Industriel (ISTDI)

- BTS PROGRAMS :
- Genie Chimique des procédés (GCP)

- Maintenance des systèmes informatique (MSI)
- Informatique Industriel (II)
- Electrotechnique (ET)
- Electronique (EN)
- Froid et climatisation (FC)
- Maintenance et après-vente automobile (MAVA)
- Bâtiment (BAT)
- Travaux publics (TP)
- Géomètre topographe (GT)
- Energie Renouvelable (ER)
- Génie Bois (GB)
- Chaudronnerie (CH)
- Fabrication Mécanique (FM)

➤ DSEP PROGRAM:

- Réseaux et Télécommunication (RT)

➤ LICENCES PROFESSIONNELLES INDUSTRIELLES in partnership with the University of DSCHANG:

- Administration des systèmes Réseaux
- Génie Logiciel
- Automatique et informatique Industrielle
- Electrotechnique
- Electronique
- Management des systèmes Automobiles
- Maintenance et Expertise des Automobiles
- Maintenance des systèmes Industriels
- Génie Energétique et Industriel
- Production et commercialisation du bois
- Ameublement et construction du bois
- Génie Mécanique et productique

➤ MASTER PROFESSIONNEL INDUSTRIEL in partnership with the University of DSCHANG

- Génie Electrique et informatique Industrielle
- Génie des Réseaux et Télécommunications
- Systèmes d'information et Génie Logiciel
- Systèmes d'information Audit et Conseil

All these schools offer courses of an international character in the sense that they are aligned with the new LMD pedagogical system, and the skills acquired by our students as well as the spirit of openness and adaptation that we teach them allow to carry out the missions entrusted to them in all the organizations and to consider all possible the possible options for their

professional future in a changing global context, and whose environment is more and more demanding.

ABSTRACT

This document presents the development of a modern community charity platform tailored for non-governmental organizations (NGOs) to enhance their operational efficiency and community engagement. The platform serves as a centralized hub for fundraising, campaign management, and donor coordination, enabling NGOs to connect more effectively with supporters. Key functionalities include secure donation processing, robust user authentication with role-based access control, and comprehensive campaign management tools that allow organizations to set goals and monitor progress. The system leverages a stack architecture using Next.js for the frontend, Prisma ORM for database management, and PostgreSQL for persistent data storage. Real-time analytics and reporting features foster transparency and accountability, which are essential for building trust with stakeholders. The platform is designed with accessibility and scalability in mind, ensuring that users can interact with the system from various devices and that NGOs can adapt the platform to their evolving needs. By streamlining core operations and providing actionable insights, this charity platform empowers organizations to maximize their social impact and strengthen their connections within the community.

Keywords: charity platform, NGOs, fundraising, campaign management, donor coordination, analytics, transparency, accessibility, Next.js, Prisma, PostgreSQL.

RESUME

Ce document présente le développement d'une plateforme caritative moderne destinée aux organisations non gouvernementales (ONG) afin d'améliorer leur efficacité opérationnelle et leur engagement communautaire. La plateforme sert de centre centralisé pour la collecte de fonds, la gestion des campagnes et la coordination des donateurs, permettant aux ONG de se connecter plus efficacement avec leurs soutiens. Les principales fonctionnalités incluent un traitement sécurisé des dons, une authentification robuste des utilisateurs avec contrôle d'accès basé sur les rôles, ainsi que des outils complets de gestion de campagnes permettant aux organisations de définir des objectifs et de suivre leur progression. Le système utilise une architecture full-stack avec Next.js pour le frontend, Prisma ORM pour la gestion de la base de données et PostgreSQL pour le stockage des données. Des fonctionnalités d'analyses et de rapports en temps réel favorisent la transparence et la responsabilité, essentielles pour instaurer la confiance avec les parties prenantes. La plateforme est conçue pour être accessible et évolutive, garantissant une utilisation sur divers appareils et une adaptation aux besoins changeants des ONG. En rationalisant les opérations clés et en fournissant des informations exploitables, cette plateforme caritative permet aux organisations de maximiser leur impact social et de renforcer leurs liens au sein de la communauté.

Mots-clés : plateforme caritative, ONG, collecte de fonds, gestion de campagnes, coordination des donateurs, analyses, transparence, accessibilité, Next.js, Prisma, PostgreSQL

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LIST OF ABBREVIATIONS

AWS	Amazon web services
CSS	Cascading style sheet
DFD	Dataflow diagram
ERD	Entity Relationship Diagram
FRQ	Functional requirements
HTML	Hypertext machine language
JWT	JSON Web Token
MFA	Multifactor Authentication
NFRQ	Non-Functional requirements
NGO	Non-governmental organization
ORM	Objected oriented mapping
RBAC	Role-based access control
RDS	Relational Database service
SNS	Simple notifications service

GENERAL INTRODUCTION

1. BACKGROUND STUDY AND CONTEXT

In the past years, philanthropy as a field has advanced in its knowledge by sharing or reaching out too many donors and volunteers.[1] This has made many organizations to change their way of collecting funds or donations from various donors. Non-governmental organizations who manage their operations from sources such as membership fees, government contracts, sales of services and goods, commercial ventures and partnerships use charity as the core source and form of support-to-support organizations around the world [2]. Charity is an important component for the organization's success. It is important for the organization's survival, expansion and development [3] non-governmental organizations (NGOs) are diverse groups of enterprises who are engaged in non-profit activities such as charity so as to offer assistance to those who are less capable. The ever-changing nature of information and communication technology has made it possible to develop many ICT-based inventions that have been able to enhance many services across many industries. Charity can be defined as the act of kindness shown by non-governmental organizations that are able to contribute part of their surplus income so as fulfil the needs of those who are less capable especially those that are affected by pandemics. The Charities Act 2011 provides the list of charity purposes as: the prevention of relief or poverty, the advancement of education, the advancement of religion, health, disability, financial hardship and many more to help those who are in need of charity. In the existing system scenario concerning charity management, non-governmental organizations such as the Institute of Charity Management are involved in collecting donations and funds from other organizations to carry out charity activities. However, there exists various challenges in the existing system. There is lack of trust from the public especially when it comes to fundraising for charity hence the rate in which people donate or pay for charity has recently reduced [4] Poor communication is also a challenge that exists in the existing system of charity management hence little communication between the organization and various donors. A lot of resources are being used in NGOs during charity hence the need to develop a charity management system to increase efficiency. This study examines the potential of a community charity platform to enhance organizational efficiency and expand outreach for charities [1] [5]. By

providing a tool to address common challenges faced by these organizations, the platform aims to improve accessibility to information for both donors and beneficiaries, ensure the security of sensitive donation data to foster trust, and minimize reliance on physical offices, thereby allowing charities to allocate more resources towards their core missions.

2. PROBLEM STATEMENT

In Cameroon, there is a gap between people who want to help their communities and the organizations that need support. Donors find it hard to find trustworthy NGOs, while NGOs struggle to reach potential donors and volunteers[6]. Without a central platform to find verified organizations, track donations, and find volunteer opportunities, it becomes difficult for communities to get help and for kind people to make a difference.

3. OBJECTIVES OF THE SYSTEM

3.1. MAIN OBJECTIVE

The main goal is to create an easy-to-use charity platform that connects donors and volunteers with verified NGOs in Cameroon through a transparent online system. This will help build trust, simplify donation tracking, and improve resource management for NGOs.

3.2. SPECIFIC OBJECTIVES

The specific goals of the platform include:

- **Investigating Challenges:** To look into the problem's charity organizations face in managing donor funding.
- **Assessing Similar Solutions:** To check how well existing systems work in the charity sector.
- **Designing the System:** To develop a system that addresses the challenges NGOs face in the community.
- **Testing the System:** To ensure the web-based system works well and is effective.

4. WORK PLAN

Table 1: Work plan

PHASE	Task description	Duration
1.Planning and Analyses	Determine project objectives, goals, and scope and requirement gathering	1 Week
2.Design	Develop a wireframe day and create a design concept	3 weeks
3.Development task	Develop the user interface components including the frontend and the backend.	7-8 Weeks
4.Testing	Develop a comprehensive testing plan and identify bugs and do testing	3 Weeks
5.Deployment	Plan and prepare for deployment.	3 Weeks

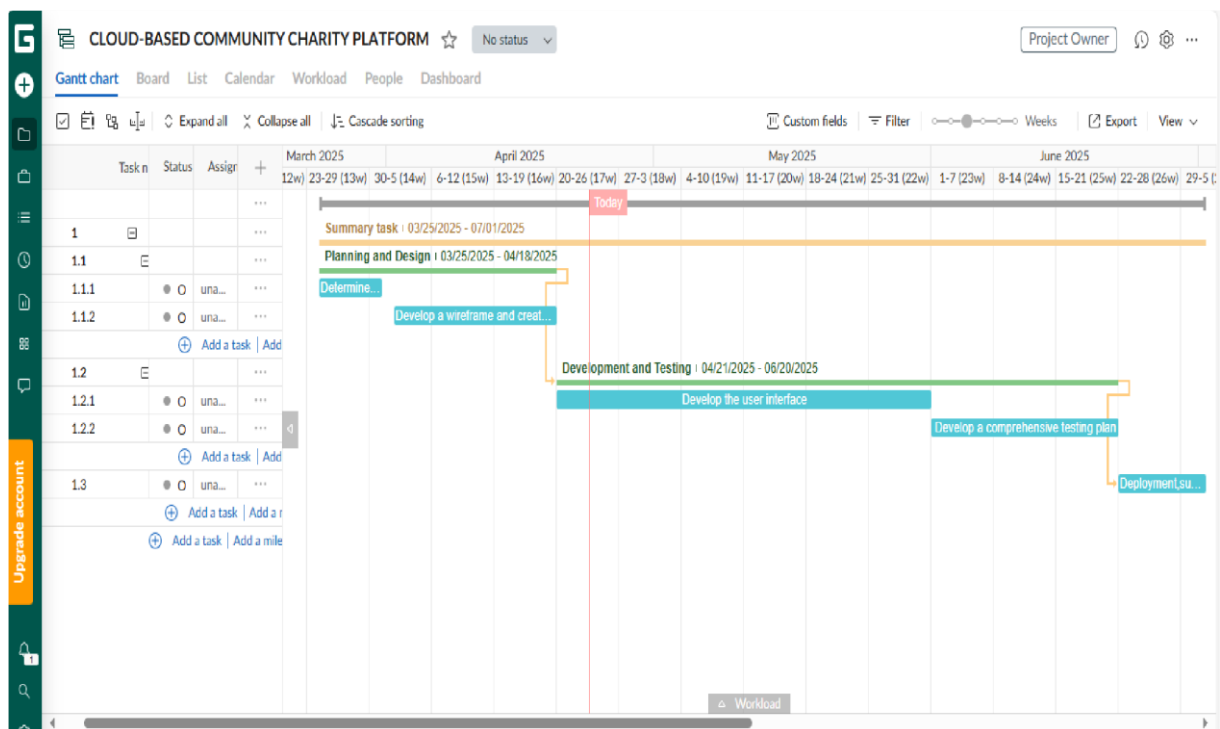


Figure 1: Chart Illustration

CHAPTER 1: LITERATURE REVIEW

1. INTRODUCTION

Charitable giving and community support have long been essential in addressing social and economic challenges, particularly in developing countries like Cameroon [7]. The rise of digital technology has led to the emergence of online charity platforms, which have transformed the way donors, volunteers, and non-governmental organizations (NGOs) interact and collaborate. These platforms, such as GoFundMe and GlobalGiving, provide tools for fundraising, campaign management, and community engagement, making it easier to mobilize resources for social causes. However, despite the global success of such platforms, many local communities in Africa still face significant barriers, including lack of trust, limited access to technology, and challenges in verifying the legitimacy of organizations [8]. This literature review examines the current landscape of charity platforms, the challenges faced in charitable giving, and the limitations of existing solutions, highlighting the need for a transparent and locally relevant platform tailored to the needs of Cameroonian communities.

2. A REVIEW OF EXISTING SYSTEMS

A review of existing charity platforms examines how online systems like GoFundMe, JustGiving, and GlobalGiving connect donors, volunteers, and organizations. It highlights their main features, benefits, and limitations to identify areas for improvement in digital charity solutions.

2.1. GOFUNDME

GoFundMe is a prominent online crowdfunding platform established in 2010, designed to facilitate fundraising for personal causes, emergencies, charities, and community projects. It empowers individuals and organizations to share their stories, connect with supporters, and raise money for a wide range of needs from medical expenses and disaster relief to educational initiatives and community improvements [9]. The platform's user-friendly interface and social media integration

have contributed to its widespread adoption, allowing users to reach a broad audience and garner support from friends, family, and even strangers. The development of GoFundMe involved several key methodologies that contributed to its success, including user-centered design to enhance usability, agile development for rapid iteration, data-driven decision-making to optimize performance, and robust security measures to protect users. Additionally, the platform emphasizes community engagement through social media integration and feedback mechanisms, fostering a supportive environment for both campaign creators and donors.

2.1.1. Advantages OF GOFUNDME

- i. **User-Friendly Interface:** The platform is easy to navigate, making it accessible for users of all ages and technical backgrounds.
- ii. **Wide Reach:** Campaigns can be shared on social media and other platforms, allowing for a broad audience and increased visibility.
- iii. **No Upfront Costs:** There are no fees to create a campaign, and funds can be withdrawn as soon as they are raised.
- iv. **Flexible Funding:** Users can choose between all-or-nothing funding or keep what they raise, allowing for flexibility in achieving their goals.
- v. **Community Support:** Many campaigns gain traction through community engagement, leading to strong support networks.
- vi. **Trust and Credibility:** GoFundMe is a well-established platform with a reputation for reliability, which can encourage potential donors to contribute.

2.1.2. Limitations of GoFundMe

- i. **Fees on Donations:** While there are no upfront costs, GoFundMe does charge a percentage fee on donations (typically around 2.9% + \$0.30 per donation), which can reduce the total amount received.
- ii. **Competition:** With millions of campaigns, it can be challenging for individual campaigns to stand out and attract donations.
- iii. **Dependence on social media:** Success often relies on the ability to promote campaigns effectively on social media, which may not be accessible to everyone.

2.2. JUST GIVING

JustGiving is a global online fundraising platform that enables individuals and charities to raise money for various causes, including medical expenses, disaster relief, education, and community projects. Launched in 2000, it has become a preferred choice for registered charities, event fundraisers, and individuals supporting social initiatives [10]. The platform allows users to create personalized fundraising pages, share their stories, and collect donations from friends, family, and the public. The development of JustGiving employs several key methodologies that contribute to its effectiveness. User-centered design is prioritized to ensure an intuitive experience for users of all technical backgrounds. Agile development practices allow for rapid updates and enhancements based on user feedback. JustGiving also utilizes data analytics to track fundraising trends and optimize campaign performance, while robust security measures protect sensitive donor information. Community engagement is fostered through social media integration, enabling fundraisers to share their campaigns widely and connect with supporters.

2.2.1 Advantages of JUSTGIVING

Here are some advantages of JustGiving;

- i. **Flexible Fundraising Options:** Users can choose from various fundraising options, including one-time donations, recurring donations, and event-based fundraising.
- ii. **User-Friendly Interface:** JustGiving's intuitive design makes it accessible to users regardless of their technological expertise.
- iii. **Wide Range of Causes:** Users can fundraise for a variety of causes, including personal, charitable, and community projects, catering to a broad audience.
- iv. **Trust and Credibility:** With a long-standing reputation, JustGiving is recognized for its reliability, which can encourage potential donors to contribute.

2.2.2 Limitations of JUSTGIVING

Here are some Limitations of JUSTGIVING;

- i. **Transaction Fees:** JustGiving charges a fee on donations (usually around 5% plus payment processing fees), which can reduce the total amount received by fundraisers.

- ii. **Competition:** The platform hosts numerous campaigns, making it challenging for individual fundraisers to stand out.
- iii. **Dependence on Promotion:** Success heavily relies on the ability to promote campaigns effectively, often requiring a strong social media presence.
- iv. **Potential for Fraud:** While JustGiving has measures to prevent fraud, the risk remains, which can impact donor trust.

2.3 GLOBALGIVING

GlobalGiving is an international crowdfunding platform that connects donors with grassroots projects and vetted nonprofits around the world. Established in 2002, it aims to support community-led organizations, particularly in developing countries, by facilitating donations for a diverse array of causes, including education, health, disaster relief, and economic development[5] [11]. The platform allows users to discover projects that resonate with them, make donations, and track the impact of their contributions. GlobalGiving employs several key methodologies to enhance its effectiveness. The platform prioritizes user-centered design, ensuring ease of navigation for donors and project creators alike. Agile development practices allow for continual improvements based on user feedback and emerging needs. GlobalGiving also utilizes rigorous vetting processes to ensure that projects meet quality and transparency standards, fostering trust among donors. Data analytics play a crucial role in tracking project performance and donor engagement, while community engagement is promoted through storytelling and impact reporting, allowing donors to see the tangible results of their contributions.

2.3.1 Advantages of GLOBALGIVING

Here are some advantages of GlobalGiving;

- i. **Focus on Grassroots Projects:** GlobalGiving emphasizes support for community-led initiatives, allowing donors to contribute to meaningful, local efforts.
- ii. **Diverse Causes:** Users can explore a wide range of projects across various sectors, including education, health, and environmental sustainability.
- iii. **Community Engagement:** The platform fosters a sense of community through storytelling and shared experiences, enhancing donor connections to projects.

- iv. **Global Reach:** GlobalGiving connects donors with projects from around the world, enabling support for international causes.
- v. **Impact Tracking:** Donors can see how their contributions are utilized through regular updates and impact reports from project leaders.

2.3.2 Limitations of GLOBALGIVING

Here are some limitations of GlobalGiving;

- i. **Transaction Fees:** GlobalGiving charges a fee on donations (typically around 5% plus payment processing fees), which reduces the net amount received by projects.
- ii. **Competition Among Projects:** With many projects available, it can be challenging for individual initiatives to capture donor attention.
- iii. **Dependence on Promotion:** Success often hinges on effective promotion and marketing, which may be a barrier for some grassroots organizations.
- iv. **Short-Term Focus:** Many projects may aim for immediate funding needs rather than addressing long-term sustainability, which can affect lasting impact.

Table 2: Literature review summary

Technology	Domain of application	Methodology	Advantages	Limitations
GoFundMe	Online crowdfunding for personal causes and charities	User-centered design, agile development, data-driven decision-making, security measures, community engagement	User-friendly interface Wide reach No upfront costs Flexible funding options Community support Trust and credibility	Fees on donations Competition among campaigns Dependence on social media Potential for fraud Limited support for nonprofits Short-term focus

Technology	Domain of application	Methodology	Advantages	Limitations
JustGiving	Global fundraising platform for individuals and charities	User-centered design, agile development, data analytics, social media integration, community engagement	Wide range of causes Integration with social media User-friendly interface Support for charities Flexible fundraising options Trust and credibility	Transaction fees Competition among projects Dependence on promotion Limited support for non-registered causes Potential for fraud Short-term focus
GlobalGiving	International crowdfunding platform connecting donors with grassroots projects	User-centered design, agile development, rigorous vetting processes, data analytics, community engagement	Focus on grassroots projects Diverse causes Impact tracking Global reach Community engagement	Transaction fees Competition among projects Dependence on promotion Limited support for non-vetted projects Potential for fraud Short-term focus

2.4 TECHNOLOGICAL APPROACH

Modern charity platforms use a range of technological solutions to improve user experience and trust. They are often built as secure web or mobile applications, integrating reliable payment gateways for safe and easy donations. These platforms implement user authentication, role management, and NGO verification systems to ensure only legitimate organizations can raise funds[12]. Transparency is enhanced through real-time reporting tools and dashboards that show how donations are used and the impact achieved. Additionally, features for managing volunteer

opportunities and strong data security measures are included to protect user information and support community engagement.

2.5 CONCEPTUAL FRAMEWORK

The conceptual framework for this Community Charity Platform is based on trust, transparency, and community engagement. It includes input variables (NGO verification, donor trust, volunteer participation), process variables (platform features, user experience, security), and output variables (successful fundraising, community impact, sustained engagement). The framework shows that when NGOs are verified and transparent, donor confidence increases, leading to higher donations and volunteer participation. The platform's technological features facilitate this process by providing tools for effective communication and trust-building, creating a sustainable ecosystem where transparency leads to trust, trust leads to engagement, and engagement leads to meaningful community impact.

2.6 CHALLENGES FACED WITH THE EXISTING SYSTEMS

Charitable giving, especially in developing countries like Cameroon, faces several significant challenges that hinder its effectiveness and reach. One of the main issues is the lack of trust and transparency, as donors often worry about whether their contributions will be used appropriately and if the organizations they support are legitimate [13]. Many NGOs struggle to verify their credibility, making it difficult for donors and volunteers to confidently engage with them. Additionally, limited access to technology and digital literacy can prevent both organizations and community members from fully utilizing online charity platforms [14]. NGOs also face difficulties in reaching a broad audience due to scarce marketing resources and the absence of a central platform for promoting their campaigns. Furthermore, donors frequently lack access to detailed progress reports or success stories, making it hard to track the real impact of their contributions. These challenges highlight the need for more transparent, accessible, and trustworthy solutions to support charitable giving in regions like Cameroon.

2.7 GAPS IN THE EXISTING SYSTEMS

In summary, while existing charity platforms have made it easier to connect donors, NGOs, and volunteers, they often fall short in areas crucial for developing regions like Cameroon. Key gaps include limited verification of NGOs, insufficient transparency in how donations are used, lack of

support for local payment methods, and minimal focus on volunteer engagement. These shortcomings highlight the need for a dedicated platform that builds trust, provides clear impact reporting, supports both donations and volunteering, and is accessible to local users addressing the unique.

3. CONCLUSION

The literature review has identified significant challenges in charitable giving, particularly in developing countries like Cameroon. Existing platforms face limitations including limited NGO verification, insufficient transparency, high fees, and lack of local relevance. These gaps create barriers to trust and effective community support. The review highlighted the importance of NGO verification, transparency, and comprehensive volunteer engagement features. These identified limitations justify the development of the Community Charity Platform, which will address these shortcomings by providing comprehensive verification, transparent reporting, volunteer management, and local payment support. The proposed platform will bridge the gap between donors, volunteers, and organizations in Cameroon, ultimately improving community impact and trust in charitable giving.

CHAPTER 2: SYSTEM ANALYSIS AND DESIGN

1. INTRODUCTION

System analysis and design of a cloud-based Community charity platform for NGOs involves a comprehensive approach to develop a robust and effective solution for connecting donors and facilitating charitable activities for non-governmental organizations. This process integrates principles from software engineering, cloud computing, and user experience design to create a secure, scalable, and user-friendly platform accessible via the internet.

2. SYSTEM ANALYSIS

System analysis cover more on what the system seeks to achieve and meet the needs of the users. It will also give more detail on the functions of different users in the system and how they are able to interact with each other on occasional basis. The chapter will also cover on how the system operates with various modules in the system and the interaction among the various modules.

2.1 FUNCTIONAL REQUIREMENTS

These are the specific features and capabilities that the system must perform, such as user registration, donation processing, NGO verification, and volunteer management.

Table 3: Functional requirements

ID	Description
FRQ 1	The system should allow Admin to sign in
FRQ 2	The system should allow Admin to login
FRQ 3	The system should allow Admin to view reports of Donor details, NGO details, NGO requests and donations.
FRQ 4	The system should allow Admin to add a user
FRQ 5	The system should allow Admin to remove NGOs and Donors.

FRQ 6	The system should allow NGO to sign in.
FRQ 7	The system should allow NGO to login.
FRQ 8	The system should allow NGO to send donation requests
FRQ 9	The system should allow NGO to receive donations
FRQ 10	The system should allow NGO to reset their password using email.
FRQ 11	The system should allow Donors to sign in
FRQ 12	The system should allow Donors to login
FRQ 13	The system should allow Donors to view requests
FRQ 14	The system should allow Donors to make donations as per the requests.
FRQ 15	The system should allow Donations to be sent to NGOs

2.2 NON-FUNCTIONAL REQUIREMENTS

These are the quality attributes and constraints of the system, such as performance, security, usability, reliability, and scalability that define how the system should behave.

Table 4: Non-Functional requirements

ID	Category	Description
NFRQ 1	Usability	The system should be easy to use
NFRQ 2	Performance	The system should perform faster and efficiently.
NFRQ 3	Reliability	The system should be reliable to all users.
NFRQ 4	Security	The system should provide efficient security to all users and their details.
NFRQ 5	Supportability	The system should support internet enables devices.
NFRQ 6	Availability	The system should be available to all users with strong internet connectivity.

3. **SYSTEM ARCHITECTURE**

The charity platform uses a modern full-stack architecture with Next.js frontend (React-based) and backend API routes. User requests flow through HTTPS to the frontend, which handles UI rendering and makes API calls to the backend for data operations. The backend uses Prisma ORM to communicate with PostgreSQL database, storing all persistent data including users, organizations, campaigns, and donations. Authentication is managed through JWT tokens with role-based access control for different user types (Admin, Donor, Organization, Volunteer). Static assets are served from public directories, and environment variables manage sensitive configuration. This layered architecture ensures security, scalability, and maintainability with clear separation between user interface, business logic, and data storage.

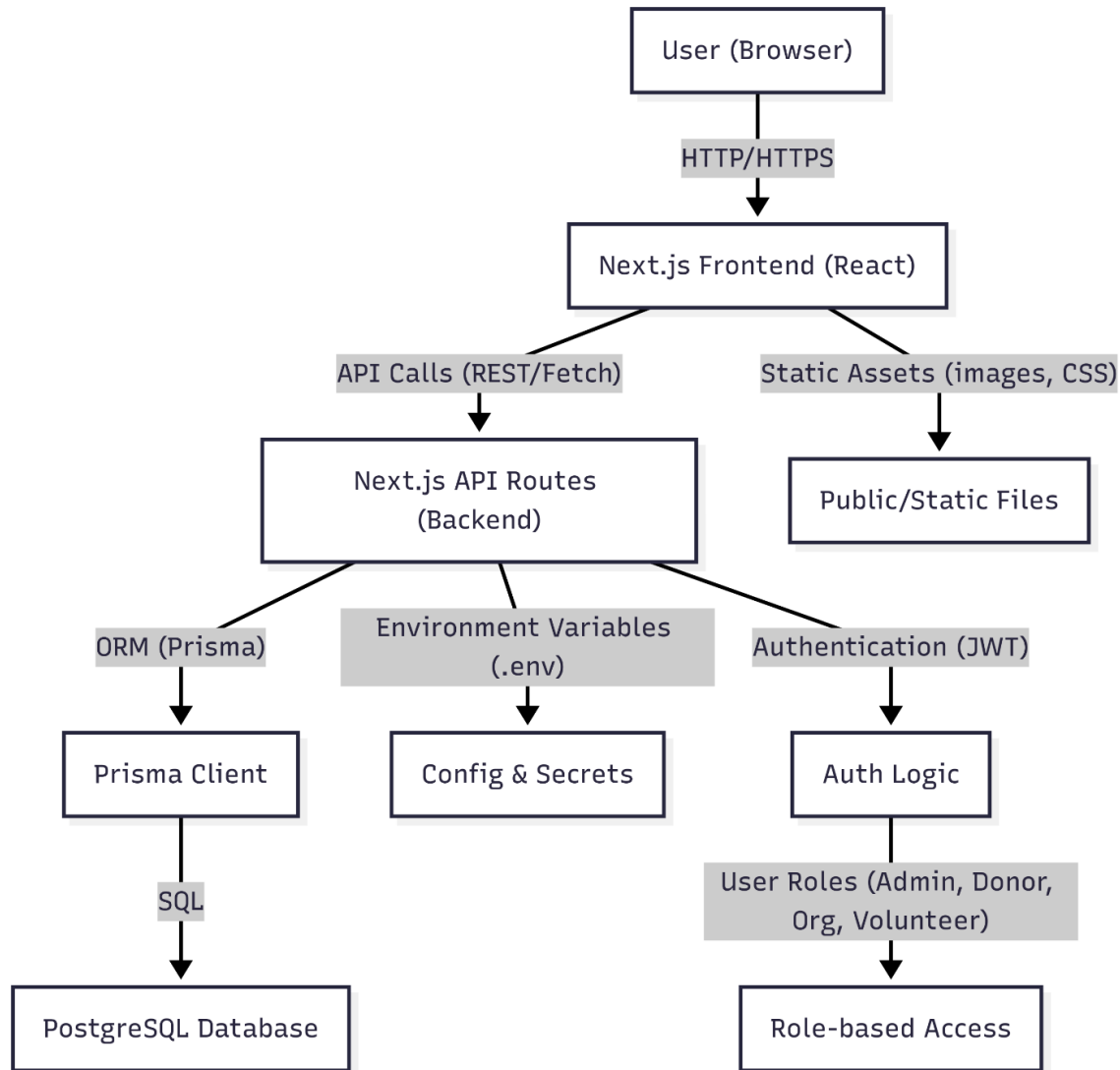


Figure 2: System Architecture

3.1 SYSTEM DEVELOPMENT METHODOLOGY

Methodology is the theoretical or systematic analysis of data or methods that are being applied in the field of study. It consists of theoretical analysis of methods and principles in the branch of knowledge. It offers theoretical understanding of a set of methods applied in a specific case [15]. Rapid Application development is the most suitable development methodology since it prioritizes quicker feedback from clients after a long period of development and testing cycles.

3.1.1 Agile Methodology

Addressing the rigidity of traditional models, the Agile approach develops software through iterative cycles (sprints), enabling continuous testing and adaptation. Unlike linear methods, Agile allows teams to incorporate feedback and refine requirements at each stage, ensuring faster delivery of functional components while maintaining flexibility for changes. The Agile model is a flexible and iterative approach to software development that emphasizes collaboration, customer feedback, and rapid delivery of functional software.

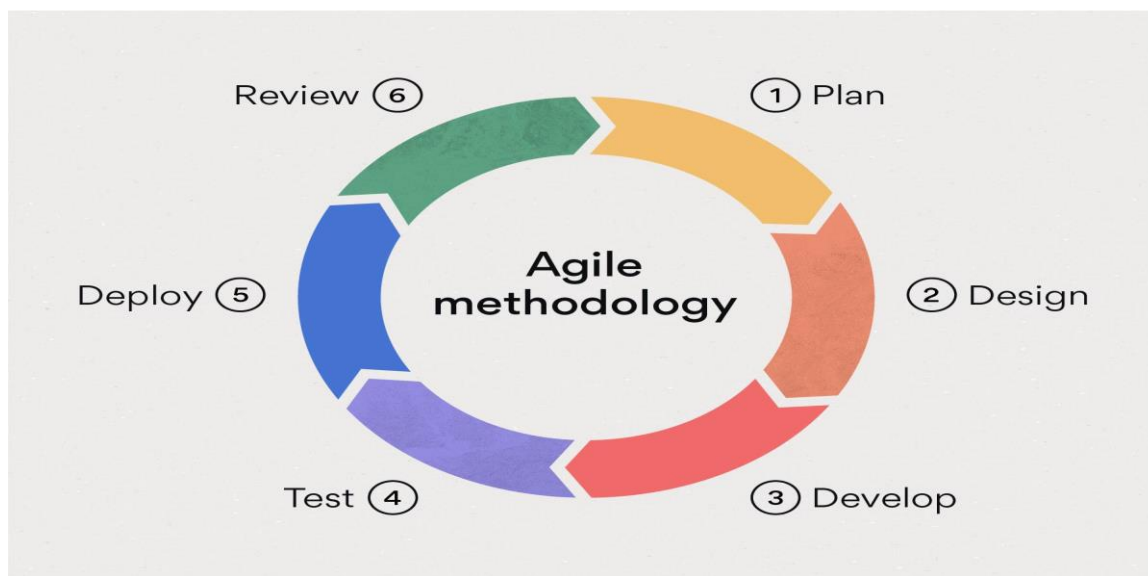


Figure 4: Agile Methodology)

1. Planning.

The Agile process begins with a crucial planning phase. During this stage, the team gathers to discuss the overarching goals of the project and establish a clear vision of what they aim to achieve. This collaborative effort results in the creation of a prioritized list of features known as the "backlog." The backlog serves as a roadmap for the project, outlining what needs to be developed and providing a reference point for the team throughout the development cycle. This phase ensures that all team members are aligned on objectives and expectations, laying a solid foundation for the work ahead.

2. Design.

Once the planning phase is complete, the team moves into the design phase. Here, they conceptualize how the software will look and function, focusing on creating a user-friendly experience. The team discusses and decides on the main features to build first, considering user needs and technical feasibility. This design blueprint is essential for guiding the development process, as it outlines the visual and functional aspects of the software. By prioritizing key features, the team ensures that they deliver the most valuable elements to users in the initial stages.

3. Development.

The development phase follows the design stage, where the team executes the plan by building the first set of features. This work is organized into short, time-boxed periods known as sprints, typically lasting one to four weeks. During each sprint, team members collaborate closely, writing code and creating the software according to the specifications established in the design phase. The focus on incremental development allows the team to make steady progress while maintaining flexibility to adapt to any changes or challenges that arise.

4. Testing.

After development, the team enters the testing phase, which is critical for ensuring the quality and reliability of the software. The team rigorously tests the newly developed features to verify that they function correctly and are free of bugs. This process may involve various testing methods, including unit testing, integration testing, and user acceptance testing. By identifying and addressing any issues before the software is released to users, the team enhances the overall user experience and builds trust in the product.

5. Deployment.

The Deployment phase follows the development stage, where the team delivers the completed product or new features to users. This process involves careful planning to determine which features will be released, often guided by user feedback and sprint results. Deployment typically utilizes Continuous Integration and Continuous Deployment (CI/CD) practices to automate testing and streamline the release process. Before going live, User Acceptance Testing (UAT) may be conducted to ensure the product meets user needs and expectations. Once deployed, the team actively gathers feedback to identify any issues and inform future updates. Additionally,

documentation and training may be provided to help users navigate new features, while ongoing monitoring and support are crucial to address any performance issues that may arise after the launch.

6. Review

The Review phase follows the deployment stage and serves as a critical opportunity for the team to reflect on the work completed and gather feedback from stakeholders. During this phase, the team conducts a formal review meeting, often referred to as a sprint review, where they present the deployed features and demonstrate how they meet the requirements outlined in previous phases. Stakeholders provide input on the product, discussing what worked well and what could be improved. This feedback is essential for identifying areas for enhancement and guiding future development efforts. Additionally, the team evaluates their performance, discussing challenges faced during the sprint and lessons learned. This reflective process fosters continuous improvement, ensuring that the team can adapt and refine their practices for future sprints, ultimately enhancing the overall quality of the product.

3.2 SYSTEM DESIGN

System design is defining the architecture, interfaces, modules and data that has to be in the system to satisfy the user requirements. It could also be seen as the applicability of systems theory on software development. Use case diagrams will be applied in the proposed system so as to identify different access and user levels in the system. The administrator is in charge of every operation that take place in the system. Other users in the system include; Donors and Non-governmental organizations NGOs. Entity Relationship Diagram ERD will be used to show the various relationships that exist among various users in the system and their roles in the system. A database schema shall also be used to observe the relationship among various modules in the system.

4. MODULES AND SYSTEM ARCHITECTURE

This describe the structure of the system and roles of all modules that are in the system.

a. Administrative module.

This module offers duties such as managing data concerning non-governmental organizations NGOs, managing all transactions that take place, managing all data containing donations that have been made and communication among charity organizations and donors. It also features approval of the organizations documents in order for the organization to be granted access into the system.

b. NGOs module.

This module features non-governmental organizations and charity-based organizations. It enables the organizations registration and uploading of legal documents to the system to request for approval to access the system. Allows requests to be sent to various donors through emails there after the donor is able to accept the request and a notification is sent concerning whether the request has been accepted or rejected. Once accepted, a donation is made.

c. Donor module.

This module only features a single repository of all information regarding donors. It includes a history of all donations that have been made to various NGOs for charity activities, communications and activities relating to charity. The module also features capturing of various transactions through credit cards, online banking, cheques and cash. Distribution of donations to organizations or charity-based events is also captured in the system.

4.1 USE CASE DIAGRAM FOR A COMMUNITY CHARITY PLATFORM

The diagram below illustrates the main actors and their interactions with the core functionalities of the Community Charity Platform. It highlights how Admins, Donors, and NGOs engage with the system through various use cases such as registration, login, campaign management, and donation processes.

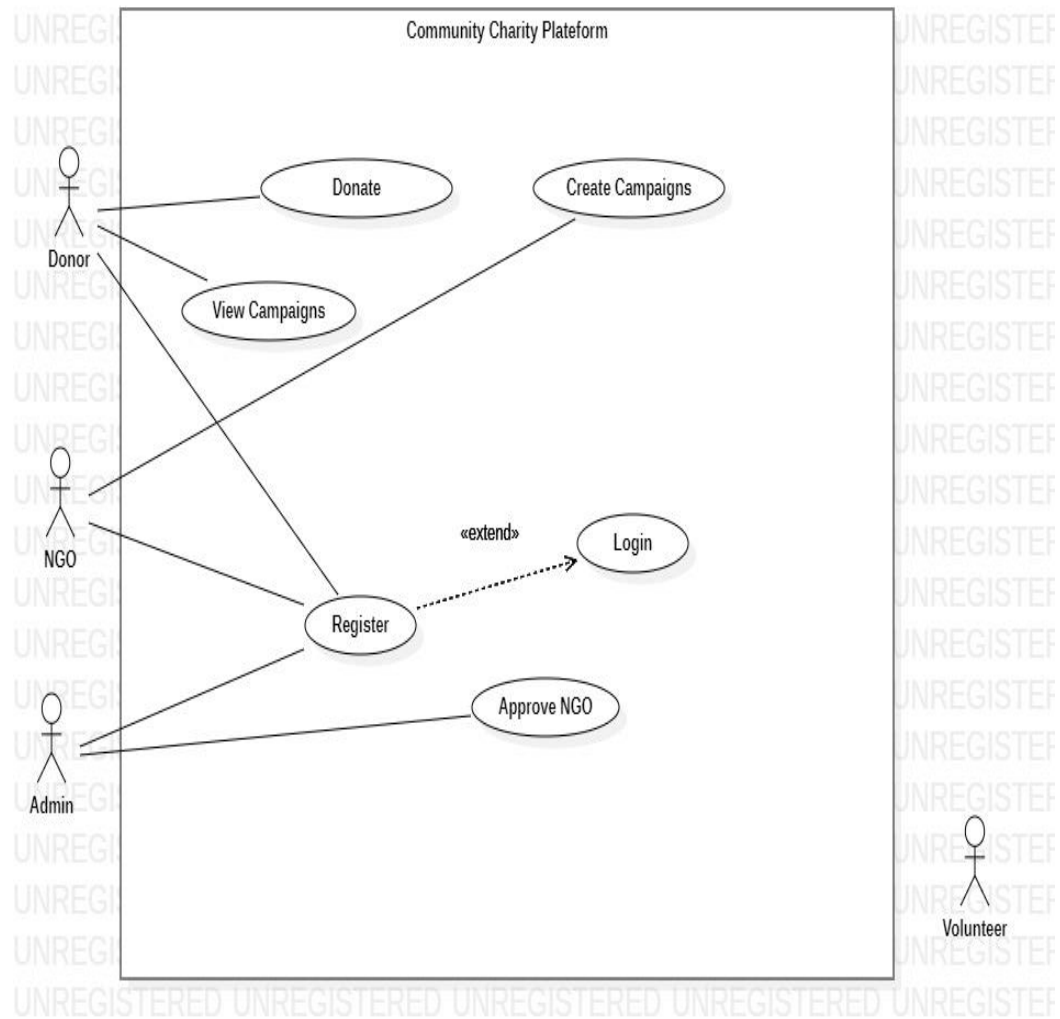


Figure 5: Use Case Diagram

Table 5: Use case analysis for system administrator

USE CASE NAME	Secure Management for Admin
ACTORS(S)	System Administrator
TRIGGER	System requires administrative oversight, user management, or platform maintenance for charity operations.
PRE-CONDITION	System administrator has active credentials with proper verification Administrative privileges are configured and active

USE CASE NAME	Secure Management for Admin
NORMAL COURSE	Authenticate with secure login and administrative access control Monitor platform performance and user activities Manage NGO account approvals and rejections
ALTERNATIVE COURSE	A1. Unauthorized Access Attempt: Triggers re-authentication and security notification Creates security incident log A2. Emergency System Maintenance: Requires dual admin verification Implements temporary system restrictions
POST-CONDITION	All administrative actions logged in immutable audit trail No direct database access without proper logging
ASSUMPTION	All administrative interfaces use HTTPS/TLS encryption No direct database access without proper logging

Table 6: Use case analysis for Donor

USE CASE NAME	Secure Management for Donors
ACTORS(S)	Donor
TRIGGER	Donor initiates a donation or updates personal information.
PRE-CONDITION	Donor has active credentials with proper verification. Donor account is registered in the system. Role-based permissions are configured.
NORMAL COURSE	Authenticate with secure login and role-based access control Select a donation campaign or project Process secure payment with encryption Generate and download donation receipts.

USE CASE NAME	Secure Management for Donors
ALTERNATIVE COURSE	A1. Unauthorized Access Attempt: Triggers re-authentication and donor notification Creates security incident log A2. Payment Processing Error: Requires donor verification for retry Implements temporary transaction limits.
POST-CONDITION	All administrative actions logged in immutable audit trail. Platform security and performance maintained
ASSUMPTION	All administrative interfaces use HTTPS encryption No direct database access without proper logging.

Table 7: Use case analysis for NGOs

USE CASE NAME	Secure Management for NGOs
ACTORS(S)	NGOs
TRIGGER	System requires resource provisioning or maintenance for NGO operations.
PRE-CONDITION	NGO has active credentials with proper verification NGO account is registered in the system Role-based permissions are configured.
NORMAL COURSE	Authenticate with secure login and role-based access control Provision encrypted storage for NGO data Enable data backup and recovery for NGO operations

USE CASE NAME	Secure Management for NGOs
ALTERNATIVE COURSE	A1. Unauthorized Access Attempt: Triggers re-authentication and NGO administrator notification Creates security incident log A2. Emergency Override: Requires dual NGO administrator verification Limits access to sensitive resources for a short window
POST-CONDITION	All changes logged in immutable audit trail Resources tagged with data sensitivity classification
ASSUMPTION	All API communications use HTTPS/TLS encryption No direct database access permitted for NGOs

4.2 DATA FLOW DIAGRAM (DFD)

A Data Flow Diagram (DFD) is a visual representation that shows how data moves through your charity platform system, illustrating the flow of information between different processes, data stores, and external entities such as donors, NGOs, and administrators. It helps identify potential bottlenecks, security vulnerabilities, and optimization opportunities in your data handling by clearly visualizing data movement, identifying security checkpoints, showing data transformation points, and documenting compliance requirements for the entire system.

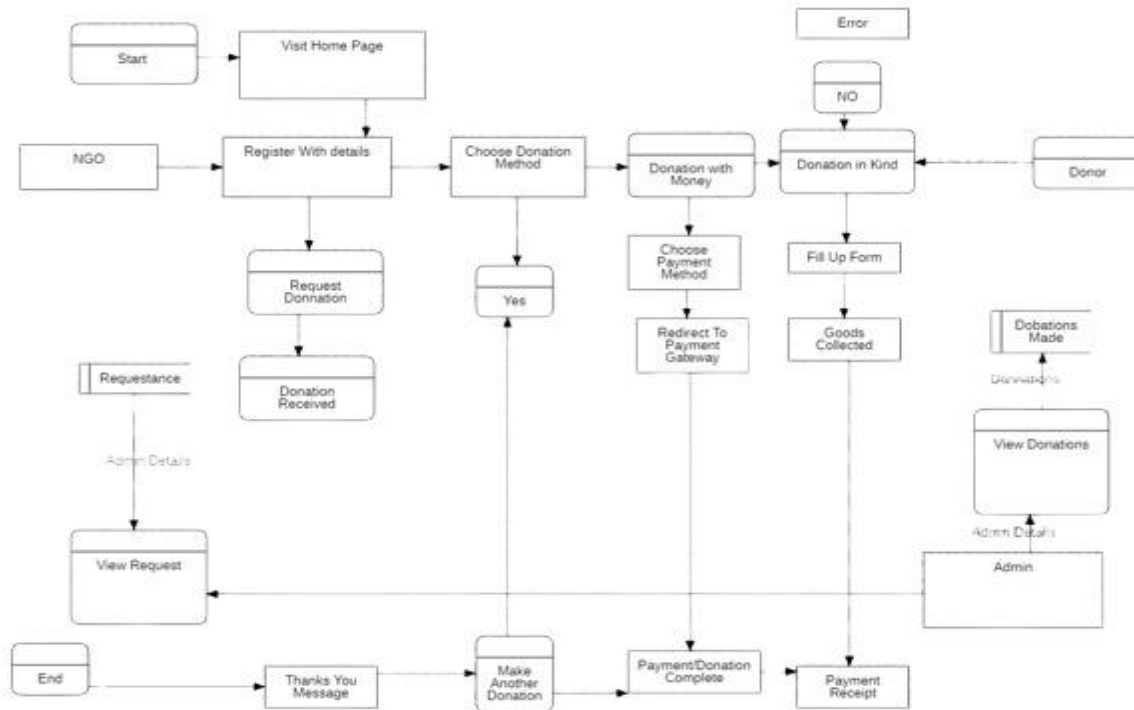


Figure 6 : Dataflow level

4.3 CLASS DIAGRAM.

A Class Diagram is a type of static structure diagram that represents the classes, attributes, methods, and relationships in an object-oriented system, such as a charity platform. It provides a visual blueprint of the system's architecture, illustrating how different classes interact with each other and detailing the attributes and operations of each class. This diagram is essential for developers and designers as it helps them understand the system's components, plan the implementation of features, and maintain a clear structure throughout the development process.

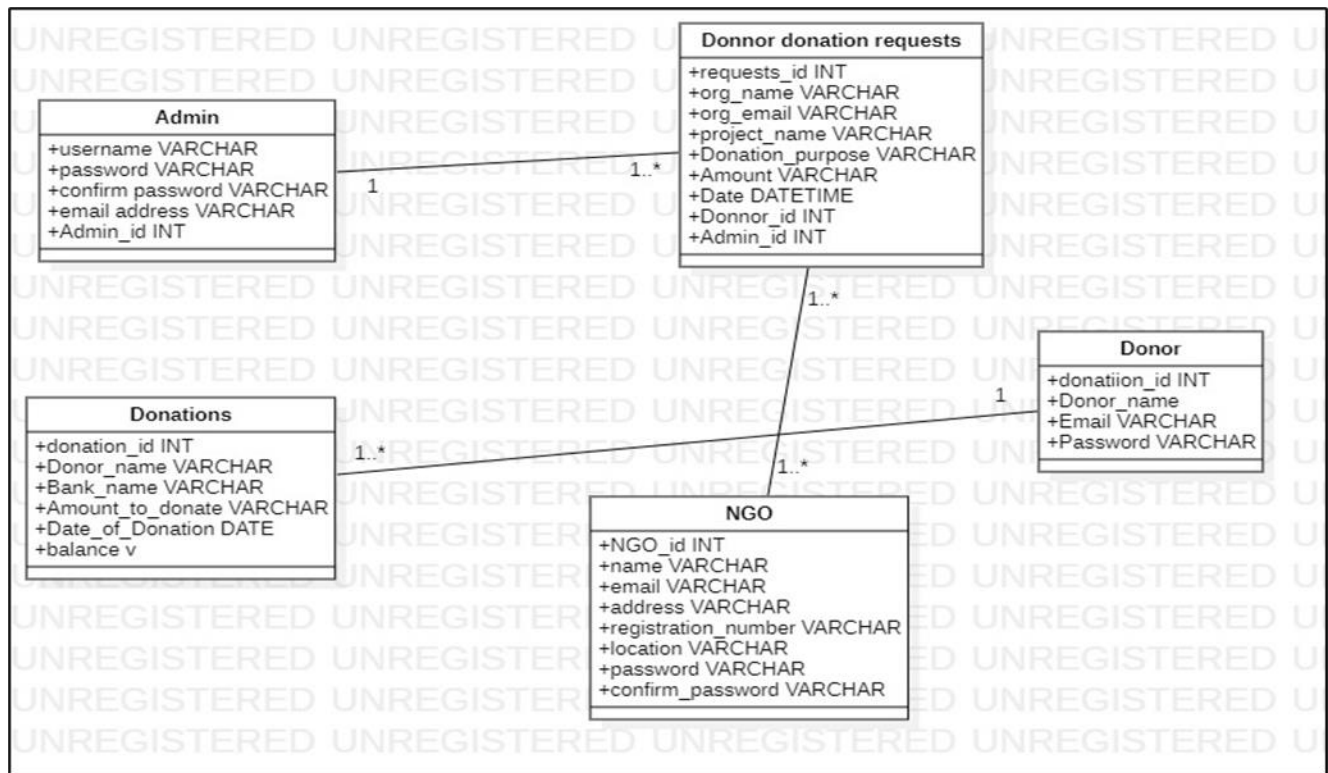


Figure 7: Database Schema

4.4 SEQUENCE DIAGRAM.

A Sequence Diagram is a type of interaction diagram that represents how objects interact in a specific scenario within an object-oriented system, such as a charity platform. It visually illustrates the sequence of messages exchanged between various components over time, detailing the order of interactions and the flow of control. This diagram is crucial for developers and designers, as it helps them understand the dynamic behavior of the system, clarifying how different classes collaborate to achieve specific functionalities. By mapping out these interactions, the Sequence Diagram aids in identifying potential issues, facilitates communication among team members, and serves as valuable documentation for future maintenance and enhancements of the system.

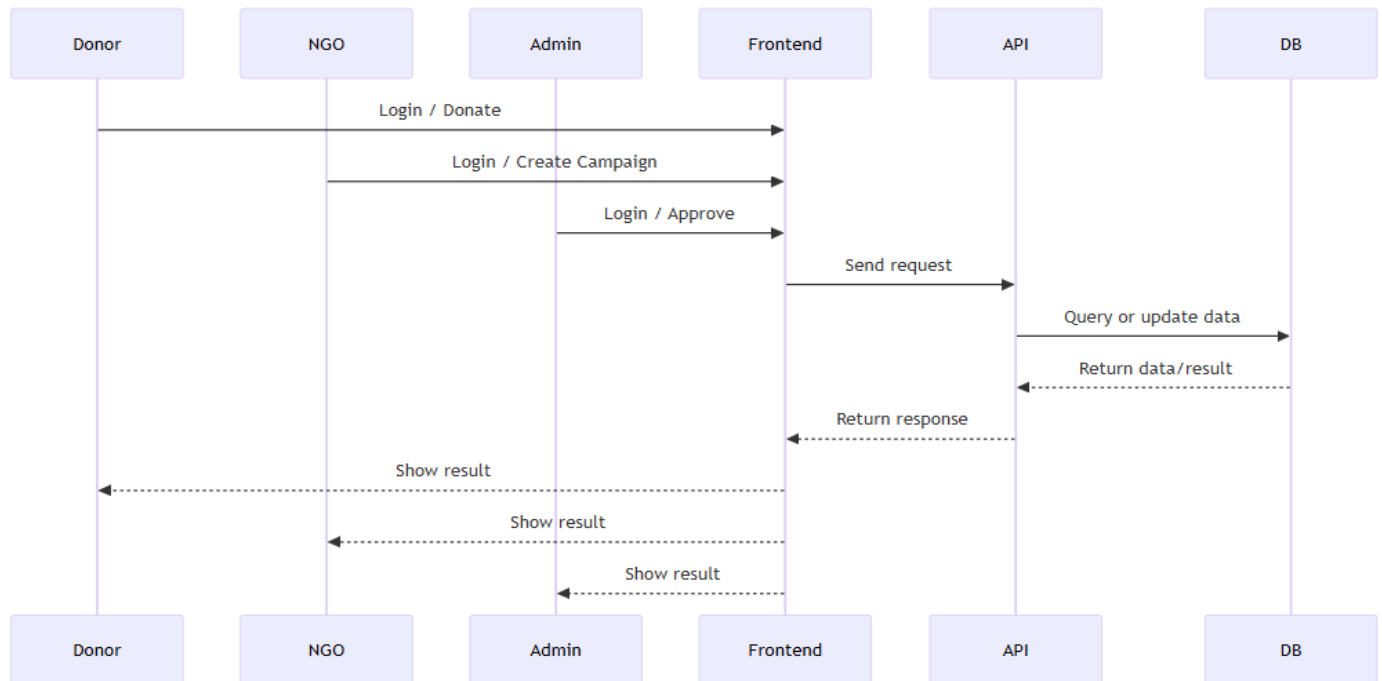


Figure 8:Sequence Diagram

5. CONCLUSION

The system analysis and design of a cloud-based community charity platform for NGOs take a comprehensive approach to meet the functional, non-functional, and business requirements of connecting donors with NGOs. By integrating various software components and leveraging cloud infrastructure, the platform facilitates efficient and transparent charitable activities. Key features include user registration and authentication for secure access, NGO management tools for easy onboarding, streamlined donation processing, and reporting functionalities that provide insights into trends and usage. The use of scalable cloud technologies ensures broad accessibility and reliability, while user-friendly interfaces help donors track their impact and NGOs monitor their fundraising performance. Overall, this platform enhances trust in charitable giving, improves donation management efficiency, and increases the impact of non-governmental organizations.

CHAPTER 3: IMPLEMENTATION AND RESULTS

1. INTRODUCTION

In the rapidly evolving landscape of digital philanthropy, the development of a robust, user-friendly charity platform is essential for connecting donors, non-governmental organizations (NGOs), and beneficiaries. This chapter details the implementation and testing phases of the charity platform, focusing on how the theoretical design was translated into a practical, fully functional web application. The primary objective was to create a reliable, secure, and efficient system that streamlines the process of campaign creation, donation management, and organizational oversight, thereby maximizing the impact of charitable activities.

2. TECHNOLOGIES, LIBRARIES AND TOOLS

The charity platform leverages a modern web technology stack, combining powerful frameworks, libraries, and tools to ensure scalability, maintainability, and a seamless user experience.

3. TECHNOLOGIES

Technology refers to the broad set of methods, practices, processes, and systems developed to solve problems and perform specific tasks. It encompasses a wide range of applications, including software, hardware and methodologies. Here's an overview of the key technologies involved:

Table 11: Technologies

Technology	Icon	Description	Version
Next.js		A React-based framework for building server-rendered and statically generated web applications. It enables both frontend and backend logic within a single codebase.	15.2.4

Typescript		A statically typed superset of JavaScript that enhances code quality and maintainability.	5.8.3
Prisma ORM		An Object-Relational Mapping tool for database management, providing type-safe database queries and migrations.	5.10.2
Node.js		The JavaScript runtime environment for executing server-side code.	22
Tailwind CSS		A utility-first CSS framework for rapid UI development and responsive design.	3.4.17

3.1 LIBRARIES

A library is a collection of pre-written code, functions, and routines that developers can use to perform common tasks without writing code from scratch. Libraries provide functionalities and abstractions to simplify the development process.

Table 8: Libraries

Technology	Icons	Description	Version
React		The core library for building user interfaces with reusable components.	18.2.0
JWT (jsonwebtoken)		Used for implementing secure authentication and authorization via JSON Web Tokens.	9.0.2
ZOD		A TypeScript-first schema validation library for validating API inputs and outputs.	3.24.1

Axios/Fetch		For making HTTP requests between the frontend and backend.	1.10.0
Prisma Client		Auto-generated client for type-safe database access.	5.10.2

3.2 TOOLS

A tool is a software application or utility designed to perform specific, often repetitive tasks to assist developers in their work. Tools can be standalone applications or integrated into larger systems to enhance productivity and efficiency. Below is an introduction to some key tools used in a community charity platform.

Table 9: Tools

Tool	Icons	Description
Visual Studio Code		The primary code editor used for development, offering rich extensions and debugging capabilities.
Prisma studio		A GUI for inspecting and editing database records during development.
Postman		Used for testing API endpoints and verifying backend logic.
Npm		Package managers for installing and managing project dependencies.

4. DATABASE DESIGN

A well-structured database is fundamental to the operation of the charity platform. The platform uses a relational database, with the schema defined and managed using Prisma ORM. The database design ensures data integrity, supports complex relationships, and enables efficient querying for analytics and reporting.

4.1 ENTITY STRUCTURE

The main entities in the database include:

- i. **User:** Stores information about all users, including their email, name, password (hashed), and role (e.g., donor, NGO, admin).
- ii. **Organization (NGO):** Contains details about registered NGOs, such as name, description, website, and the user who manages the organization.
- iii. **Campaign:** Represents fundraising campaigns created by NGOs, including title, description, target amount, current amount raised, status, and associated organization.
- iv. **Donation:** Records each donation made by users, including the amount, message, campaign, and donor.
- v. **Volunteer:** Stores information about volunteers and their participation in campaigns or organizations.

4.2 RELATIONSHIPS

The database design of the charity platform defines clear relationships between its core entities to ensure data integrity and support the platform's functionality:

- i. **User Roles:** Each user is assigned a specific role admin, donor, or organization which determines their permissions and accessible features within the system.
- ii. **Organization Management:** An organization (NGO) is linked to a user who acts as its manager, establishing accountability and enabling role-based access to organization-specific features.
- iii. **Campaign Ownership:** Campaigns are created and managed by organizations. Each campaign is associated with a single organization but can receive multiple donations from different users.

- iv. **Donation Tracking:** Every donation is linked to both the user who made the donation (the donor) and the campaign it supports, allowing for accurate tracking of contributions and campaign progress.

4.3 DATA MANAGEMENT

Data is collected through user interactions with the platform, such as registration forms, campaign creation, and donation processes. Validation is enforced at both the frontend and backend to ensure data accuracy and consistency. Prisma ORM provides type safety and schema enforcement, while Prisma Studio allows for easy inspection and management of data during development and testing.

5. DEPLOYMENT

Deployment is a critical phase that transitions the platform from development to production. The charity platform was initially deployed in a local development environment for testing and validation. The following steps were involved:

5.1 LOCAL DEPLOYMENT

The application was run locally using Next.js's development server. Prisma Studio was used to interact with the database.

Production Deployment: For production, the platform can be deployed on cloud platforms such as Vercel, AWS, or Digital Ocean. Environment variables and database credentials are securely managed to ensure data protection. AWS Amplify was used to deploy my work in the cloud. To interact with the live prototype of the application, see the link provided in Appendices.

5.2 CONTINUOUS INTEGRATION

Git and GitHub Actions (or similar CI/CD tools) was used to automate testing and deployment workflows, ensuring that updates are reliably delivered. For access to the complete source code please refer to the GitHub repository in Appendices.

6. IMPLEMENTATION DETAILS

The implementation phase involved translating the design into a working system, integrating both frontend and backend components.

6.1 BACKEND DEVELOPMENT

API routes were created under the `app/api/` directory, handling user authentication, NGO registration, campaign management, and donation processing. Prisma ORM was used for all database interactions, ensuring type safety and efficient querying.

6.2 FRONTEND DEVELOPMENT

The user interface was built using Next.js pages and React components. Tailwind CSS was used for styling, ensuring a responsive and visually appealing design. Key pages include registration, login, dashboards for different user roles, campaign listings, and donation forms.

6.3 AUTHENTICATION

JWT-based authentication was implemented to secure user sessions and protect sensitive routes. Middleware functions were used to enforce role-based access control.

Testing: API endpoints were tested using Postman, while the frontend was tested interactively in the browser. Prisma Studio facilitated database inspection and manual data entry during development.

6.4 API KEY ENDPOINTS

1) Create campaigns

Endpoint: POST `/api/campaigns`

The **POST `/api/campaigns`** endpoint allows authenticated NGOs to create new fundraising campaigns by submitting essential details like title, description, target amount, and duration. This feature empowers NGOs to efficiently initiate fundraising efforts while ensuring security through required authentication.

Request Body Example:

```
{
  "title": "Clean Water for All",
  "description": "Providing clean water to rural communities.",
  "targetAmount": 5000,
  "endDate": "2025-12-31",
  "image": "/images/water-campaign.jpg"
}
```

Figure 9: Request body

Reponse Example:

```
{
  "title": "Clean Water for All",
  "description": "Providing clean water to rural communities.",
  "targetAmount": 5000,
  "endDate": "2025-12-31",
  "image": "/images/water-campaign.jpg"
}
```

Figure 10: Reponse

Code:

```
export async function POST(request: Request) {
  // ...authentication logic...
  const { title, description, targetAmount, endDate, image } =
  const campaign = await prisma.campaign.create({
    data: { title, description, targetAmount, endDate: new Date
  });
  return NextResponse.json(campaign);
}
```

Figure 11: Code

6.5 ALGORITHMS AND DATA STRUCTURE USED

1) Campaign Progress Calculation:

- **Algorithm:** The progress of each fundraising campaign is calculated as a percentage using the formula:

$$\text{progress} = \left(\frac{\text{currentAmount}}{\text{targetAmount}} \right) \times 100$$

- **Implementation:** This calculation is performed in the backend and the result is sent to the frontend for display. To ensure accuracy, checks are implemented to handle cases where the target amount might be zero, preventing division by zero errors.

2) Data Structures

- **Campaign Object:** Each campaign is represented as an object with fields such as:
 - `id`: Unique identifier for the campaign.
 - `title`: Name of the campaign.
 - `description`: Details about the campaign.
 - `currentAmount`: Amount raised so far.
 - `targetAmount`: Total goal for the campaign.
 - `startDate` and `endDate`: Duration of the campaign.
- **User Roles:** User roles (e.g., "admin", "ngo", "donor") are stored as a string field in the user model and used to control access to different parts of the platform. A role-based access control (RBAC) system is implemented to enforce permissions based on these roles.

3) Authentication Algorithm

- **JWT Verification:** The backend uses the HMAC SHA-256 algorithm to sign and verify JSON Web Tokens (JWT) for secure authentication. When a user logs in, a token is generated and sent to the client. For protected API routes, the backend decodes and verifies the token to ensure the request is authenticated. Additionally, token expiration is managed to enhance security, requiring users to refresh their tokens periodically.

4) Data Pagination

- **Algorithm:** To efficiently handle large datasets (e.g., campaigns, donations), the backend implements pagination using the `skip` and `take` parameters in Prisma queries. This allows the system to fetch only a subset of records at a time, improving performance and user experience. Furthermore, pagination includes total count metadata to inform users about the total number of records and the number of pages available.

5) **Error Handling and Logging:**

- **Algorithm:** Implement centralized error handling mechanisms to capture and log exceptions throughout the application. This includes structured logging to provide context about errors, such as user IDs, timestamps, and request details, facilitating debugging and monitoring.

7. **RESULTS AND TESTING**

The results and testing phase focused on thoroughly evaluating the platform's functionality, performance, and user experience. Comprehensive testing was conducted to ensure that all features operated as intended and that the system met the needs of its various user roles, including admins, NGOs, and donors.

7.1 **RESULTS OBTAINED**

- i. **User Registration and Login:** Users can register and log in, with their roles determining access to specific features.
- ii. **NGO Management:** NGOs can register and submit their profiles for admin approval.
- iii. **Campaign Creation and Donation:** Approved NGOs can create campaigns, and donors can browse and contribute to campaigns.
- iv. **Donation Tracking:** Donors can view their donation history, and NGOs can track received donations.
- v. **Admin Analytics:** Admins have access to dashboards displaying key metrics such as total donations, active campaigns, and user engagement.

7.2 **USER MODULES**

The following are the modules in the system.

a) Registration page

The registration page allows new users to create an account on the charity platform by providing essential information such as name, email, and password. It ensures secure onboarding and verifies user details before granting access to platform features.

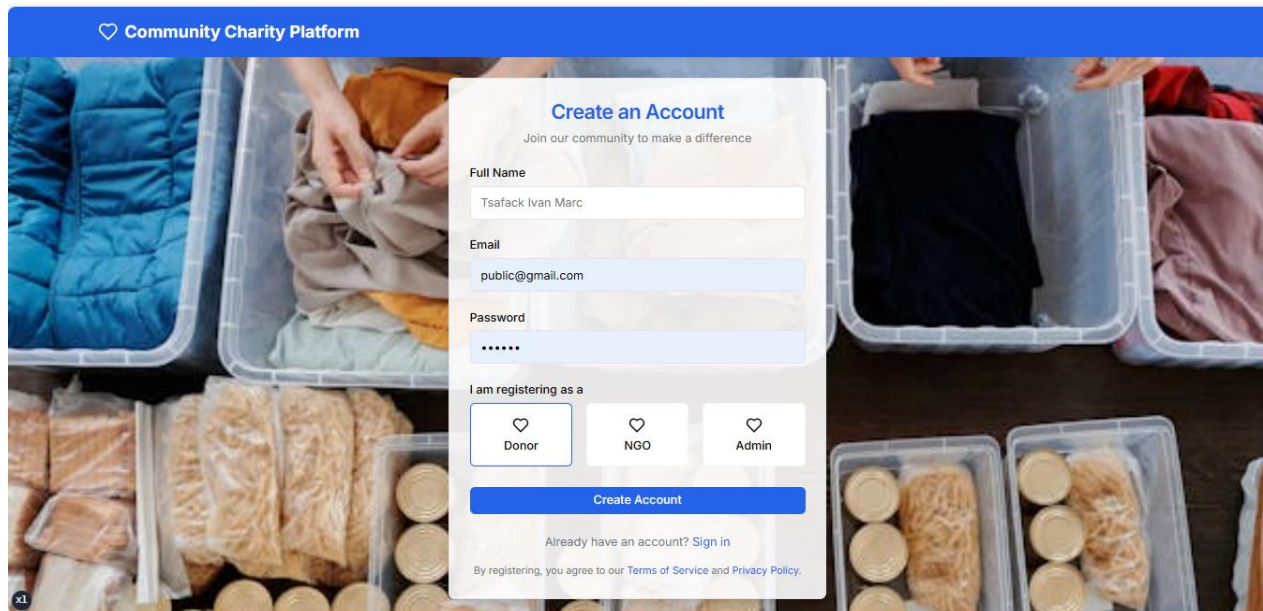


Figure 12: Registration page

b) Admin Dashboard

The admin dashboard provides system administrators with comprehensive oversight of the charity platform, including user management, NGO approvals, and platform analytics.

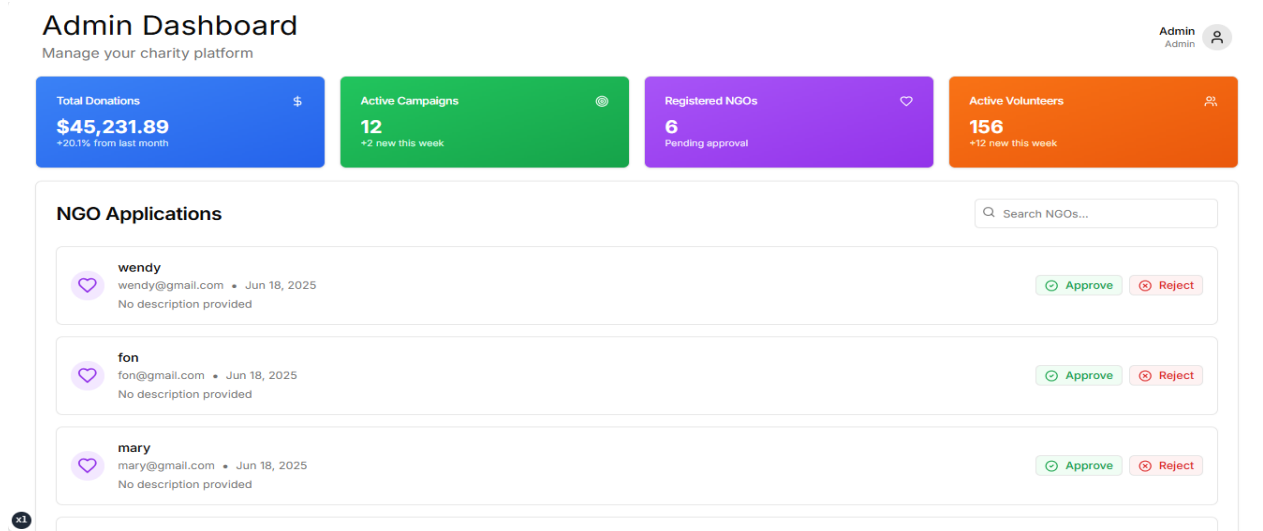


Figure 13: Admin dashboard

c) Donor Dashboard

The donor dashboard provides donors with a personalized interface to view their donation history, track campaign progress, and manage their charitable giving activities. It enables donors to explore new campaigns, monitor their impact, and access donation receipts while maintaining secure access to their personal information and transaction records.

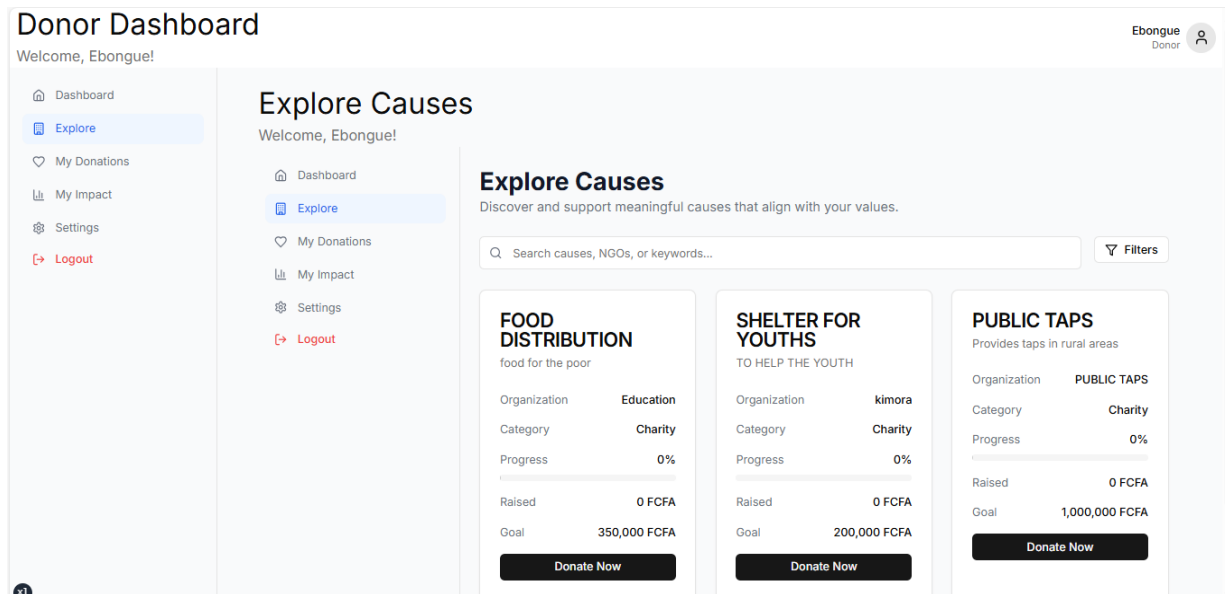


Figure 14: Donor dashboard

d) NGO Dashboard

The NGO dashboard provides registered organizations with tools to create and manage fundraising campaigns, track donations, and monitor campaign performance. It enables NGOs to update their profiles, view donor analytics, and communicate with supporters while maintaining secure access to their organizational data and campaign management features.

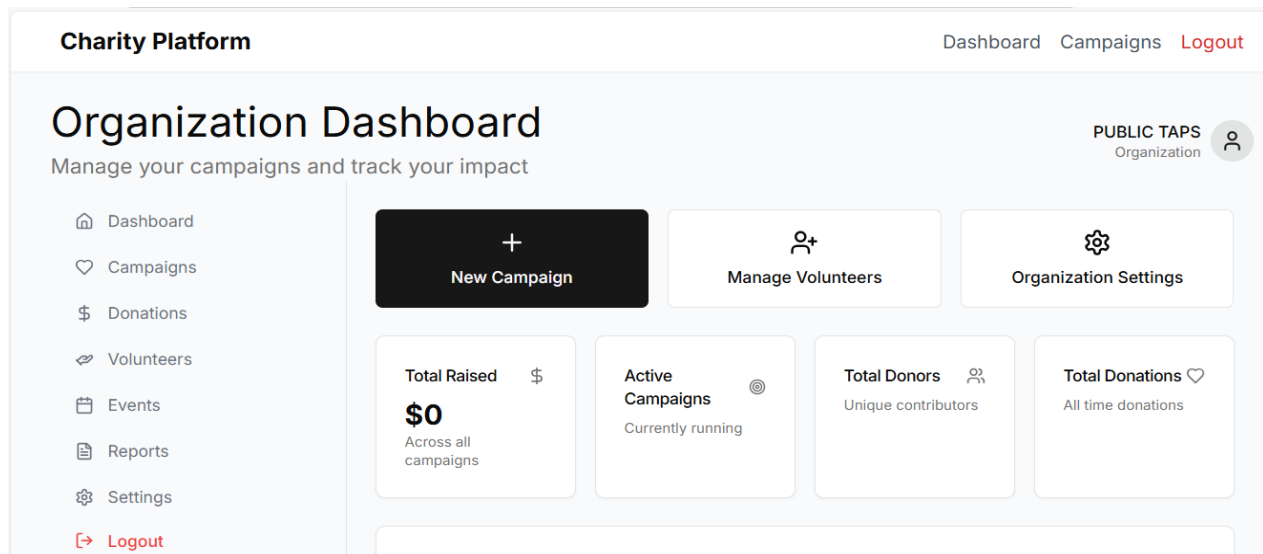


Figure 15: NGO dashboard

7.3 TESTING

Testing was a critical phase in the development of the charity platform, aimed at ensuring the system's reliability, security, and usability for all user roles. The testing process was comprehensive and covered both functional and non-functional aspects of the application.

7.3.1 Test Data

To simulate real-world usage, a variety of sample data was created and used throughout the testing process. This included:

- i. **Sample Users:** Accounts with different roles (admin, donor, NGO) were created to verify role-based access and permissions.
- ii. **NGOs:** Multiple NGO profiles were registered, some pending approval and others approved, to test the registration, approval, and management workflows.
- iii. **Campaigns:** NGOs created several fundraising campaigns with varying goals, categories, and statuses to test campaign creation, listing, and management features.
- iv. **Donations:** Donor accounts were used to make donations to different campaigns, testing the donation process, transaction recording, and dashboard updates.

7.3.2 Testing Metrics

The effectiveness of the platform was evaluated using the following functional metrics:

- i. **Successful Registration Rate:** The percentage of users and NGOs able to register and activate their accounts without errors.
- ii. **Role-Based Access Control:** Verification that users could only access features and pages permitted by their assigned roles (e.g., donors could not access admin or NGO management pages).
- iii. **Donation Processing Accuracy:** Ensuring that all donations were correctly processed, recorded in the database, and reflected in both donor and NGO dashboards.
- iv. **Data Integrity:** Confirming that all data (users, NGOs, campaigns, donations) was accurately stored, retrieved, and displayed, with no loss or corruption.
- v. **Error Rate:** Monitoring the frequency and types of errors encountered, such as failed registrations, invalid logins, or unsuccessful donations.

7.3.3 Testing Procedure

A systematic approach was adopted to test each feature and workflow of the platform:

- i. **Unit Testing:** Individual components and API endpoints were tested in isolation to ensure they performed their intended functions correctly.
- ii. **Integration Testing:** The interaction between different modules (e.g., user registration and login, campaign creation and donation) was tested to ensure seamless end-to-end workflows.
- iii. **Edge Case Testing:** Special attention was given to edge cases, such as:
 - Submitting forms with missing or invalid data (e.g., empty required fields, invalid email formats).
 - Attempting unauthorized actions (e.g., donors trying to access admin features, unapproved NGOs trying to create campaigns).
 - Handling duplicate entries (e.g., registering with an already used email).
- iv. **Manual Testing:** The user interface was manually tested across different devices and browsers to ensure responsiveness, accessibility, and a smooth user experience.

- v. **Error Handling:** The system's ability to gracefully handle errors was tested by intentionally triggering failures (e.g., network interruptions, invalid API requests) and verifying that appropriate error messages were displayed to users.

Table 10: Test case

Test case	Test Scenario	Expected outcome	Actual outcome	Status
1	User registration	User account created, redirected to login	Success	pass
2	Ngo registration	NGO profile created, pending admin approval	success	pass
3	Admin approves NGO	NGO status updated to approved	success	pass
4	NGO creates campaign	Campaign appears in dashboard and public listing	success	pass
5	Donors makes donations	Donation recorded, reflected in dashboards	success	pass
6	Invalid login credential	Error message displayed	success	pass

8. CONCLUSION

In conclusion, the implementation and testing of the charity platform have resulted in a robust, secure, and user-friendly system that streamlines the process of charitable giving and organizational management. By leveraging modern web technologies, the platform successfully connects donors, NGOs, and administrators, facilitating transparent and efficient charitable activities. The system's modular architecture, comprehensive testing, and intuitive user interface ensure that it meets the needs of all stakeholders and is well-positioned for future enhancements and scalability.

GENERAL CONCLUSION

In conclusion, the development of a digital charity platform represents a significant advancement in the way charitable activities are managed and delivered in the modern era. By leveraging state-of-the-art web technologies such as Next.js, TypeScript, Prisma ORM, and Tailwind CSS, we have demonstrated the ability to connect donors, NGOs, and administrators through a secure, scalable, and user-friendly platform. Our system addresses the limitations of traditional charity management, which often suffers from inefficiencies, lack of transparency, and limited reach. Through real-time campaign management, transparent donation tracking, and robust role-based access control, the platform empowers organizations to maximize their impact and fosters trust among all stakeholders. The integration of modern authentication, responsive design, and comprehensive analytics underscores our commitment to building a solution that is both effective and accessible. By providing NGOs and donors with actionable insights and streamlined workflows, our platform not only enhances the management of charitable activities but also contributes to greater social good and community development. Looking forward, ongoing research and development will focus on expanding the platform's features, such as integrating additional payment gateways, enhancing analytics, and supporting mobile accessibility. Ultimately, our goal is to empower organizations and individuals to make informed, impactful contributions, thereby strengthening the culture of giving and support in society.

1. RECALL OF WORK

Throughout this project, our primary focus was on designing and implementing a comprehensive charity platform that facilitates seamless interaction between donors, NGOs, and administrators. The objective was to create a reliable, efficient, and transparent system for managing registrations, campaigns, donations, and organizational oversight. This was achieved through a systematic approach involving requirements analysis, architectural design, database modeling, frontend and backend development, and thorough testing.

2. ACHIEVEMENTS

During the course of this project, several key milestones were accomplished. Firstly, we successfully designed and implemented a modular, scalable web application that supports user registration, NGO management, campaign creation, and donation processing. The use of modern frameworks and libraries ensured a robust and maintainable codebase. Secondly, we implemented secure authentication and role-based access control, ensuring that users could only access features appropriate to their roles. Thirdly, we developed intuitive dashboards and analytics features, providing stakeholders with clear insights into

platform activity and impact. Lastly, we prioritized user experience and accessibility, resulting in a platform that is easy to use and visually appealing across devices.

3. RECALL

While significant progress has been made, there remain areas for further enhancement and exploration. Future work will focus on:

- **Feature Expansion:** Adding support for additional payment gateways, volunteer management, and advanced reporting tools.
- **Mobile Accessibility:** Developing a mobile application or optimizing the platform further for mobile devices to increase accessibility.
- **Integration with External Services:** Exploring integration with third-party services for identity verification, payment processing, and social media sharing.
- **Enhanced Security:** Continuously improving security measures to protect user data and prevent fraud.
- **User Feedback and Iterative Improvement:** Collecting feedback from real users and stakeholders to guide ongoing development and ensure the platform remains relevant and effective.
- **Scalability and Performance:** Preparing the platform for increased user load and expanding its reach to support more organizations and donors.

While the charity platform has already demonstrated its potential to transform charitable giving and management, ongoing development and collaboration will be essential to further enhance its

capabilities, usability, and impact. This continuous improvement will not only benefit NGOs and donors but also contribute to building stronger, more resilient communities in a rapidly changing world.

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APPENDICES

This appendix provides access to the working prototype and source code for the project.

1) GitHub Repository

This repository contains the complete source code, detailed documentation, and additional resources related to the project.

Link: <https://github.com/tsafack-ivan/communityplatform.git>

2) Cloud Prototype

This link directs you to the live prototype hosted in the cloud, allowing you to interact with the application and test its features.

Link: <https://main.du2mbxp5ztuwq.amplifyapp.com/>